GMI German-Malaysian Institute

DIPLOMA PROGRAMME

MINI PROJECT

Academic Period: JANUARY 2026

Program	:	**DIPLOMA IN CYBER SECURITY**
Course	:	**INTRUSION DETECTION AND PREVENTION SYSTEMS**
Code	:	**CIT 0592**
Total Marks	:	**110**
Title	:	

GENERAL INSTRUCTIONS:

1. Write down clearly (with capital letters) your name, trade, group and the date of examination at the bottom of this page.
2. Do not open this question booklet until you are told to do so by the invigilator.
3. Read the specific instructions and questions carefully before answering.

This paper consists of **5** printed pages, including this page.

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Semester/Trade/Group	:	**SEM 4 / DCBS / CBS 1**

Date : **18/05/2026**

No	Assessment Criteria	CLO	PO	Level of Diff	Full Marks	Score	Marks	Comment
1.	Demonstration	1	2	C3	40			
2.	Interpreting Findings (Analyses)	1	2	C3	30			
3.	Fluency & Clarity	1	2	C2	15			
4.	Defence Ability	1	2	C2	20			
5.	Report Content & Format	1	2	C1	5			
				Total	110			
				Total Marks	/110		/30	
Total			**PO 2**		/30			
Signature								

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NO.	CRITERIA	APPROACHES EXPECTATIONS M[1]	MEETS EXPECTATIONS M[2]	EXCELLENT M[3]	WEIGHT	MARKS
ASSIGNMENT (1) – 10%						
1	**Demonstration of tools**	Able to slightly demo the demonstration details	Able to demo most of the demonstration's details	Able to demo all of the demonstration's details	M/3*40	
2	**Interpreting Findings** (Analyses/concept knowledge)	Able to slightly interpret the findings and results	Able to interpret most of the findings and results	Able to interpret all of the findings and results	M/3*30	
3	**Fluency and Clarity** Presentation	Presentation not fluent and voice is not clear.	Presentation slightly fluent and voices are slightly easy to be heard clearly.	Very fluent, and voices are easy to be heard clearly.	M/3*15	
4	**Defence ability/ presentation**	Not able to answer questions, not prepared and no confidence at all	Able to answer questions but with little preparation and confidence	Able to all answer questions very well and confidently. Very well prepared	M/3*20	
5	**Report Content & Format**	Report is poorly presented (unclear and confusing)	The report is clearly presented, but not enough.	The report is clearly presented and in details	M/3*5	
	TOTAL				**110**	

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1. Find 1 tool to simulate an attack.

2. Find 1 IDS/IPS/Firewall/Honeypot tool (run in any OS / platform: Linux platform/windows/desktop/server) -opensource

You should simulate the attack and show how you use IDS/IPS/Firewall/Honeypot to detect the attack

3. Requirement to use these tools (tool for attack/ids)

4. Procedure of installation

5. Discuss the use of these tools

6. Discuss about advantage /disadvantage of these tools

7. Explain examples on how to use the tool, explain how it works.

8. You should come out with report & discuss how to analyze the report

(information of the report)

9. Presentation (last week of the class: will be determine later)

10. Should include screen shots/ sample of reporting.

11. Submit: softcopy of the report, slide, video presentation. (BEFORE PRESENTATION)

13. For softcopy report: Front Page – Use the Cover Page provided.

(Tittle: IDS mini project-(Semester/Group)/ Current year)

create folder (name of the project)

14: Presentation slide: MAXIMUM 20 pages (include front page)

15: Produce video presentation with duration MAXIMUM 5 minutes to describe the tool as whole, from installation, simulation with attack and produce report (Submission only).

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Selected Tools

Attack Tool: ApacheBench (Web Request Flood)

Defense Tool: Wazuh (An open-source security monitoring platform)

Victim Tool: Apache2, Web Server (Running simple website)

Simulation Scenario

We will simulate an HTTP Web Flood (DDoS) attack where the attacker (kali linux) sends a massive volume of web requests to the victim's web server (ubuntu victim machine) using a load-testing tool. This is done to overwhelm the server's resources and exhaust its ability to handle legitimate traffic. Wazuh will then detect the unusual traffic in the web logs and generate a Level 10 alert because of the high frequency of HTTP error codes originating from a single source IP address.

Requirements

Operating System: Ubuntu Server 22.04 LTS & Kali Linux 2025.4

Hardware:

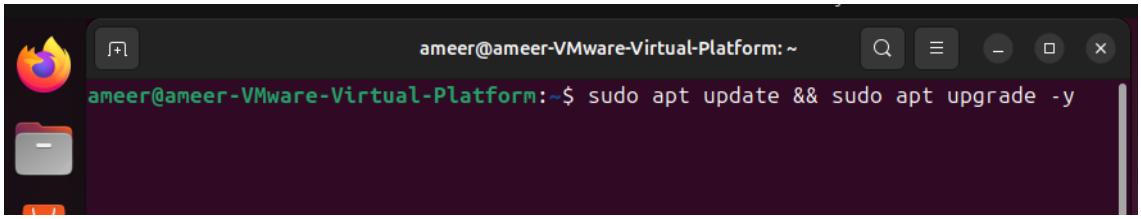
- **Ubuntu Server (Wazuh):** 100GB Disk, 10GB RAM, 4 CPU’s
- **Ubuntu Server (Victim):** 20GB Disk, 4GB RAM, 2 CPU’s.
- **Kali Linux (Attacker):** 80GB Disk, 7GB RAM, 4 CPU’s - Prebuilt VM

Networking: All machines must be on the same network (Bridge or NAT with static IPs).

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Installation Procedure

Preparing Ubuntu Machine



Updating Ubuntu to the latest version

Installing Wazuh in Ubuntu

```
ameer@ameer-VMware-Virtual-Platform:~$ sudo apt install curl -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  libwoff1
Use 'sudo apt autoremove' to remove it.
The following NEW packages will be installed:
  curl
0 upgraded, 1 newly installed, 0 to remove and 0 not upgraded.
Need to get 227 kB of archives.
After this operation, 535 kB of additional disk space will be used.
Get:1 http://my.archive.ubuntu.com/ubuntu noble-updates/main amd64 curl amd64 8.
5.0-2ubuntu10.9 [227 kB]
Fetched 227 kB in 0s (461 kB/s)
Selecting previously unselected package curl.
(Reading database ... 192669 files and directories currently installed.)
Preparing to unpack .../curl_8.5.0-2ubuntu10.9_amd64.deb ...
Unpacking curl (8.5.0-2ubuntu10.9) ...
Setting up curl (8.5.0-2ubuntu10.9) ...
Processing triggers for man-db (2.12.0-4build2) ...

Progress: [ 80%] [################################################.....]
```


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```
root@ameer-VMware-Virtual-Platform:~# curl -sO https://packages.wazuh.com/4.7/wazuh-install.sh
```

```
root@ameer-VMware-Virtual-Platform:~# bash ./wazuh-install.sh -a -i
14/05/2026 18:53:31 INFO: Starting Wazuh installation assistant. Wazuh version: 4.7.5
14/05/2026 18:53:31 INFO: Verbose logging redirected to /var/log/wazuh-install.log
14/05/2026 18:53:56 INFO: --- Dependencies ----
14/05/2026 18:53:56 INFO: Installing gawk.
14/05/2026 18:54:15 WARNING: Hardware and system checks ignored.
14/05/2026 18:54:15 INFO: Wazuh web interface port will be 443.
14/05/2026 18:54:36 INFO: --- Dependencies ----
14/05/2026 18:54:36 INFO: Installing apt-transport-https.
14/05/2026 18:54:59 INFO: Wazuh repository added.
14/05/2026 18:54:59 INFO: --- Configuration files ---
14/05/2026 18:54:59 INFO: Generating configuration files.
14/05/2026 18:55:07 INFO: Created wazuh-install-files.tar. It contains the Wazuh cluster key, certificates, and password
s necessary for installation.
14/05/2026 18:55:10 INFO: --- Wazuh indexer ---
14/05/2026 18:55:10 INFO: Starting Wazuh indexer installation.
```

Wait until the installation of Wazuh server, indexer and dashboard finish.

```
14/05/2026 19:23:53 INFO: Wazuh dashboard web application initialized.
14/05/2026 19:23:53 INFO: --- Summary ---
14/05/2026 19:23:53 INFO: You can access the web interface https://<wazuh-dashboard-ip>:443
    User: admin
    Password: ntIlgBOgEUG4uSE.tkT*XLlXzMnBgTJb
14/05/2026 19:23:53 INFO: --- Dependencies ----
14/05/2026 19:23:53 INFO: Removing gawk.
14/05/2026 19:24:12 INFO: Installation finished.
```

The credentials for the dashboard access is given after the installation is successful.

Install Attack Tool on Kali Linux (ApacheBench)

```
┌──(kali㉿kali)-[~/Echo]
└─$ sudo apt install apache2-utils -y
[sudo] password for kali:
Upgrading:
  apache2  apache2-bin  apache2-data  apache2-utils

Summary:
  Upgrading: 4, Installing: 0, Removing: 0, Not Upgrading: 2002
  Download size: 2,014 kB
  Space needed: 2,048 B / 61.3 GB available
```


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Discussion of Tools

- **ApacheBench (ab)**: Originally designed to measure HTTP web server performance, ApacheBench can rapidly generate massive numbers of concurrent web requests. It is used for load testing, benchmarking server response times and simulating HTTP flood DDoS attacks to test server resilience.
- **Wazuh:** An open-source security platform that unifies XDR (Extended Detection and Response) and SIEM (Security Information and Event Management) capabilities. It is used for log data analysis, intrusion detection, file integrity monitoring and providing real-time alerts against active threats like DDoS attacks.

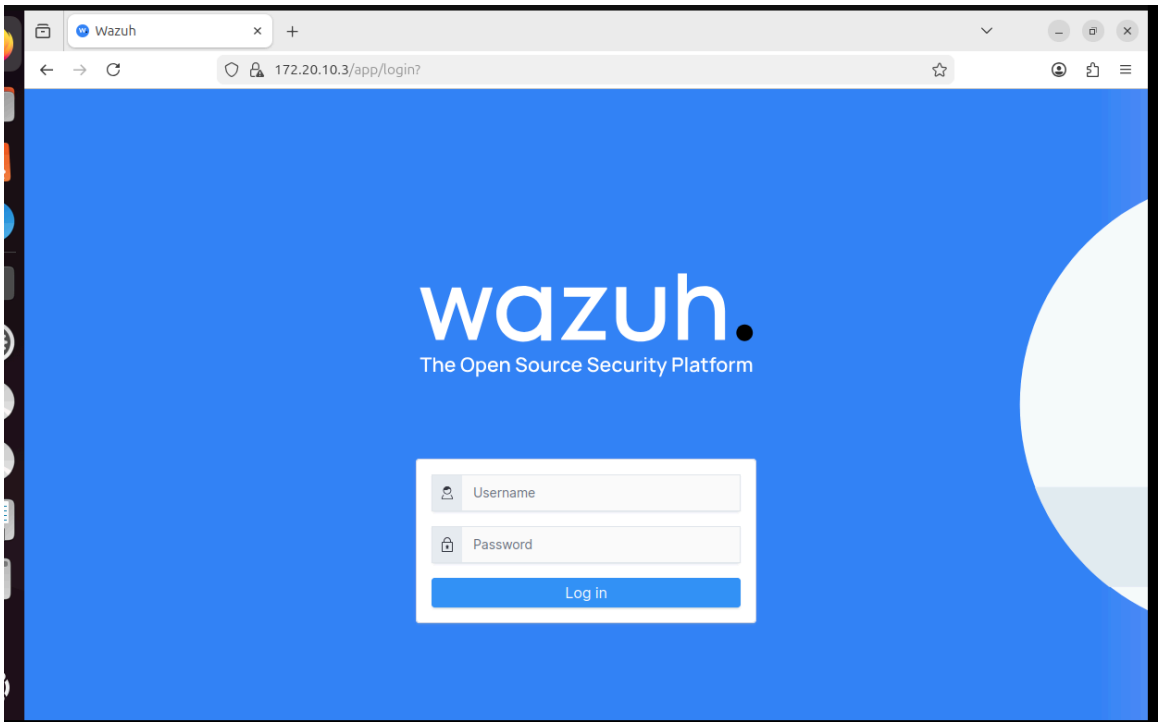
Advantages & Disadvantages

Tool	Advantage	Disadvantage
ApacheBench (ab)	Lightweight and straightforward to use; excellent for rapidly generating massive volumes of concurrent HTTP requests.	Limited to Layer 7 (HTTP/HTTPS) testing; lacks advanced evasion features like IP spoofing or complex packet crafting.
Wazuh	Comprehensive unified platform for SIEM and XDR; highly customizable rule engine for real-time threat alerting.	initial configuration and rule syntax can be complex; high resource and disk space requirements for log storage.

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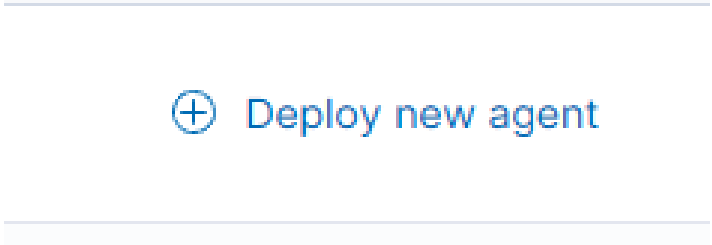
How to use the tools

Access the Wazuh Dashboard

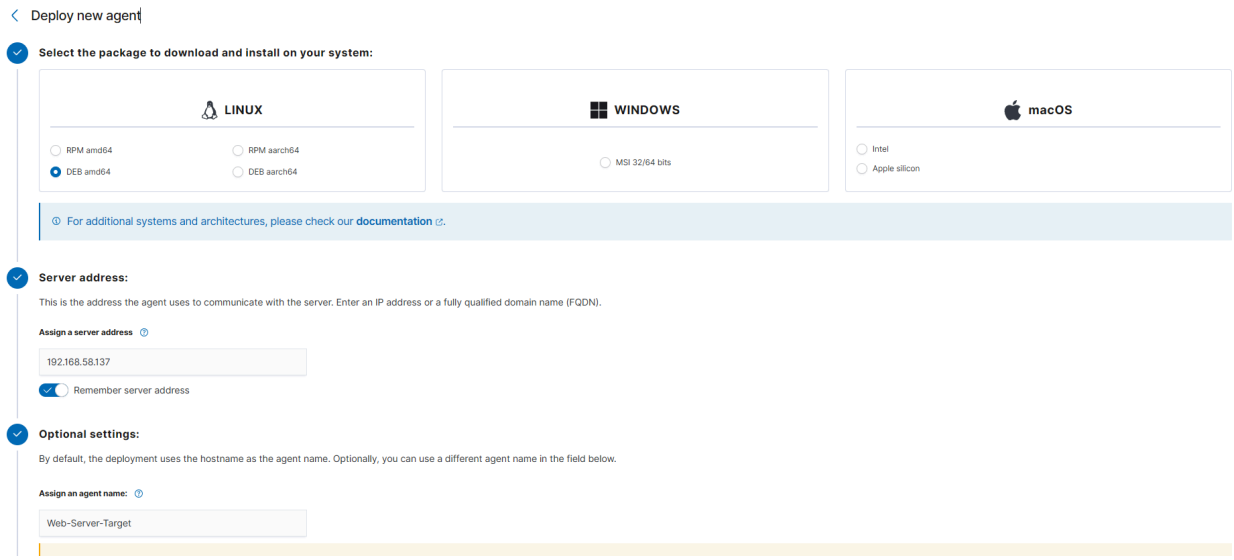


Insert the given credentials that you got during the installation.

To connect the Web Server VM (Victim), we must first deploy a new agent in the Wazuh dashboard to establish a connection.



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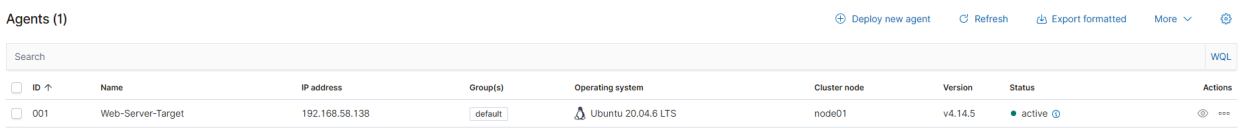
On the "deploy new agent" tab, choose the correct installation on the system, for our Victim VM (Ubuntu) we must choose Linux AMD64 architecture type. Either RPM or DEB filetype is completely optional, in this case we are going to use DEB to install the package. Next, point the server address to the Wazuh VM IP Address so the agent can communicate with the Wazuh Manager VM. Lastly, it is optional but it is better to assign a name on the Agent.



"wget https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.14.5-1_amd64.deb && sudo WAZUH_MANAGER='192.168.58.137' WAZUH_AGENT_NAME='Web-Server-Target' dpkg -i ./wazuh-agent_4.14.5-1_amd64.deb "

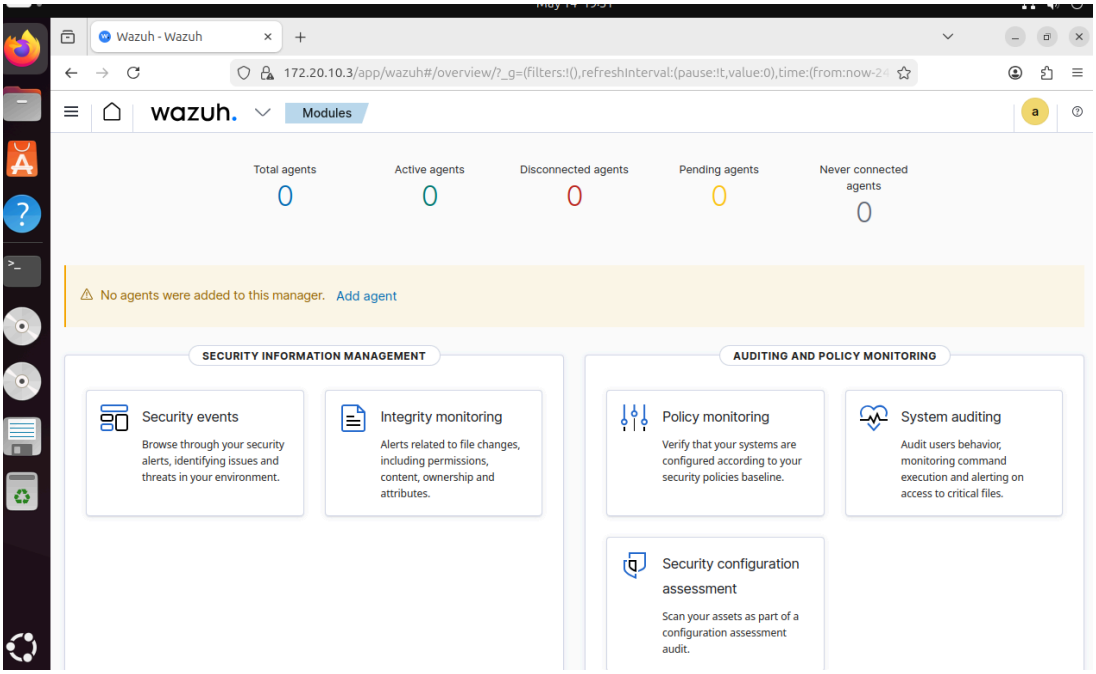
Once finished setting up the Agent, it will automatically give a wget commands to install the Agent onto the Web Server/Victim VM. Simply run it on the VM.



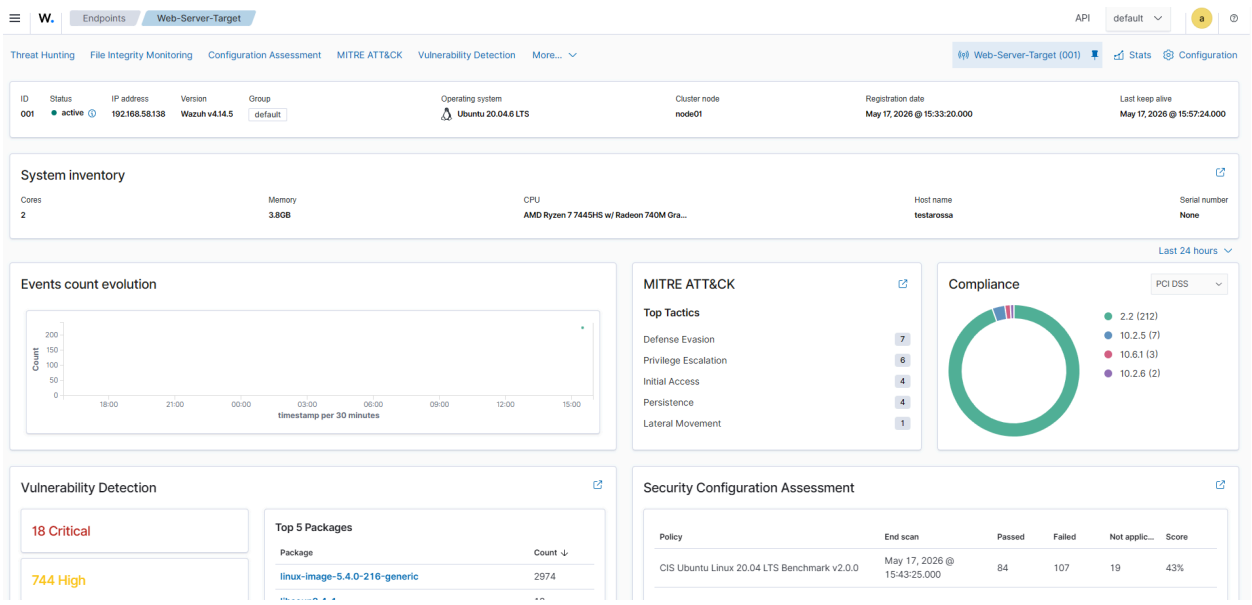


And then you'll have the Agent on the VM up and running, ready to monitor the Victim VM.

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Go to security events for the dashboard of alerts on upcoming attacks.

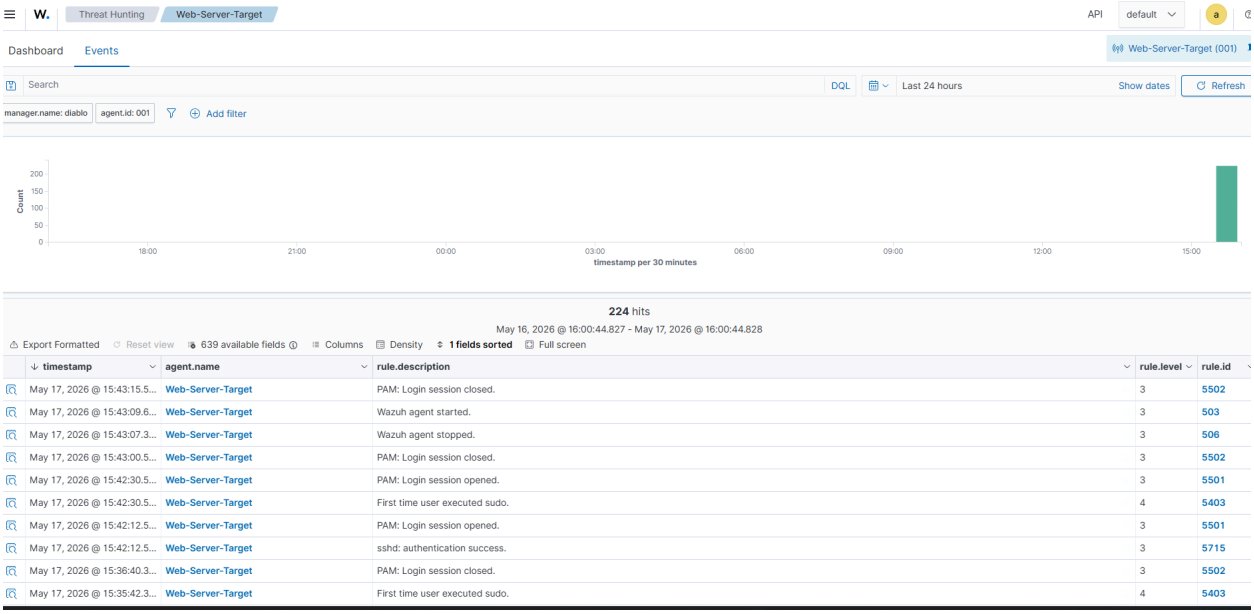


The connected Victim VM can be viewed on the Agent dashboard. Here it will display all the alerts and vulnerabilities that it finds in the machine.

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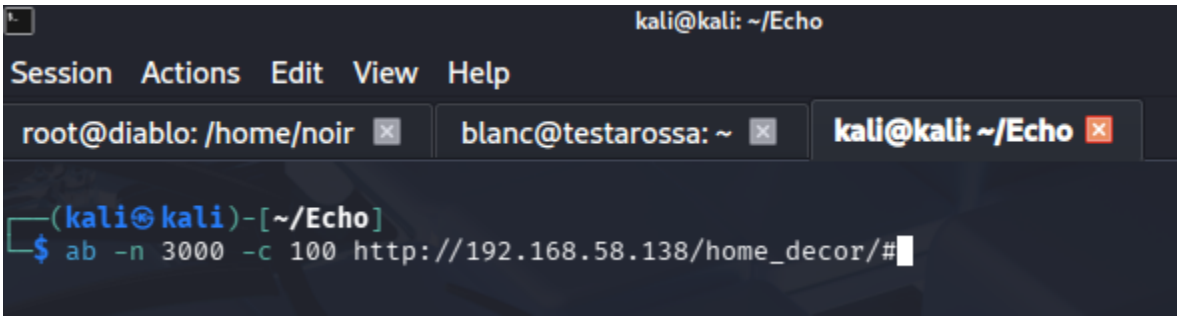
Other than Alerts, Wazuh can also detect the security vulnerabilities that are detected on the connected machine to the Wazuh VM using Wazuh Agent.



We can monitor alerts in the Events tab (Threat Hunting -> Events).

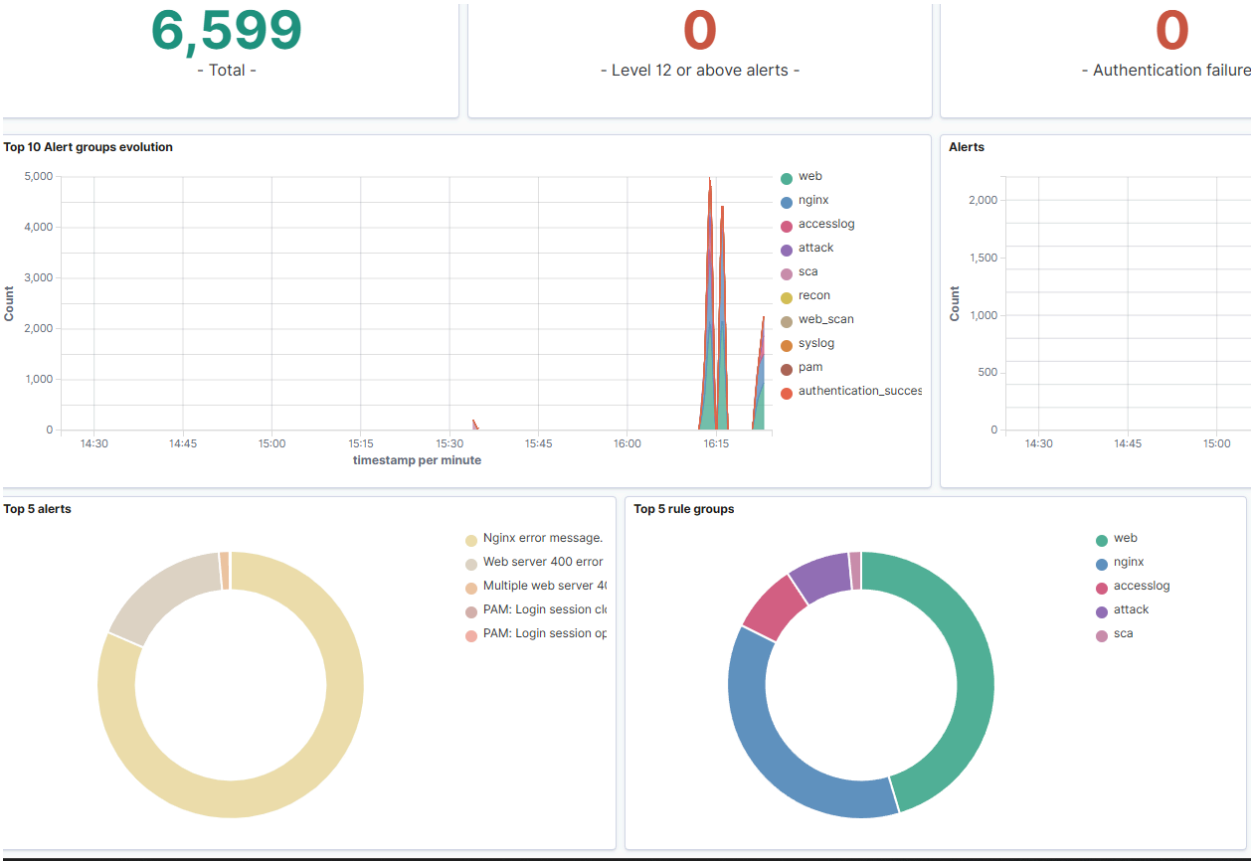
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Detecting Attack Using Wazuh

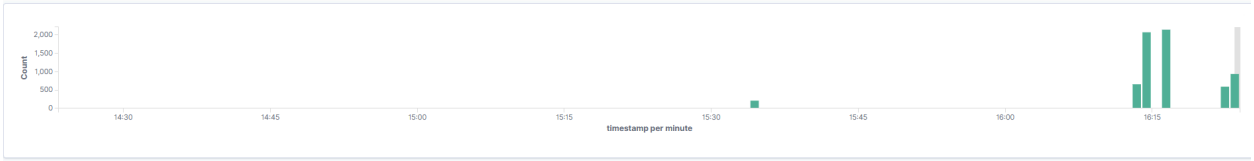


Preparing for the attacks we are going to the ApacheBench command to fire a sudden burst of 3,000 total web requests, pushing 100 at an exact millisecond interval (-c 100 stands for concurrency)

Monitor the logs, there 6,599 hits web server 400 error code, the web flood was successful with ApacheBench.



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6,599 hits

May 17, 2026 @ 14:23:24.047 - May 17, 2026 @ 16:23:24.047

Export Formatted | Reset view | 639 available fields | Columns | Density | **1 fields sorted** | Full screen

timestamp	agent.name	rule.description	rule.level	rule.id
May 17, 2026 @ 16:23:02.3...	Web-Server-Target	Web server 400 error code.	5	31101
May 17, 2026 @ 16:23:02.3...	Web-Server-Target	Web server 400 error code.	5	31101
May 17, 2026 @ 16:23:02.3...	Web-Server-Target	Web server 400 error code.	5	31101
May 17, 2026 @ 16:23:02.3...	Web-Server-Target	Web server 400 error code.	5	31101
May 17, 2026 @ 16:23:02.3...	Web-Server-Target	Web server 400 error code.	5	31101
May 17, 2026 @ 16:23:02.3...	Web-Server-Target	Web server 400 error code.	5	31101
May 17, 2026 @ 16:23:02.3...	Web-Server-Target	Web server 400 error code.	5	31101
May 17, 2026 @ 16:23:02.3...	Web-Server-Target	Multiple web server 400 error codes from same source ip.	10	31151
May 17, 2026 @ 16:23:02.3...	Web-Server-Target	Web server 400 error code.	5	31101
May 17, 2026 @ 16:23:02.3...	Web-Server-Target	Web server 400 error code.	5	31101

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Conclusion

This simulation successfully demonstrated both the execution and the detection of an HTTP Web Flood attack. By utilizing ApacheBench, we were able to simulate a volumetric DDoS event designed to overwhelm a target web server's resources. Conversely, the successful configuration of the Wazuh agent and manager showcased the critical role of real-time log monitoring and SIEM capabilities in a defensive posture. Wazuh was able to instantly ingest the Nginx access logs, identify the abnormal spike in HTTP requests, and automatically trigger a high-severity alert. Ultimately, this exercise highlights the necessity of active threat detection systems in identifying destructive network attacks early, enabling security teams to respond swiftly and mitigate potential service disruptions.

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Reference

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